MATH 201: Lecture 5a Handout

4.1 Normal Distirbution Continued

Name: Solutions Date:

Learning goals for today

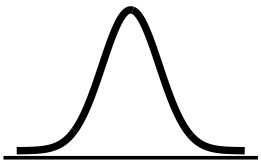
By the end of this lecture, you should be able to:

- Explain how z-scores standardize values.
- Interpret area under a normal curve as probability.
- Calculate left-tail, right-tail, and between probabilities.
- Identify and interpret unusually large or small values.
- Translate probability statements into statistical context.

Quick Review: Normal Distribution

A normal distribution is:

- Symmetric
- Bell-shaped
- Defined by mean (μ) and standard deviation (σ)
- Total area under curve = 1



We can approximate probabilities using:

- Empirical Rule (68–95–99.7)
- Or calculate them more precisely using z-scores (covered today!)

If $X \sim N(\mu, \sigma)$, what is a Z-score?

$$Z = \frac{x - \mu}{\sigma}$$

notation

# of standard deviations from the mean

If ACT scores follow a normal distribution with mean 21 and standard deviation 6:

- Compute and interpret the Z-score for someone that scores a 24.

$$z = \frac{24-21}{6} = \frac{3}{6} = 0.5$$

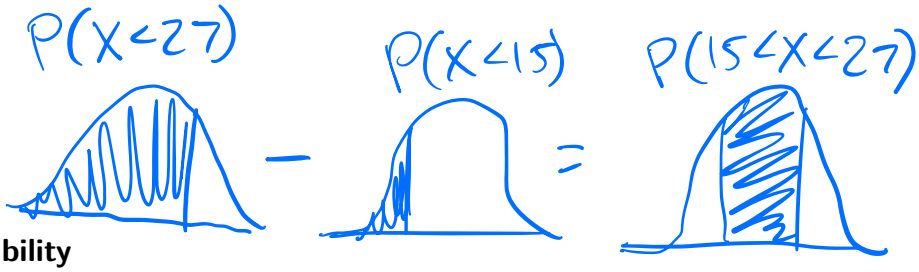
So a score of 24 on the ACT is 1/2 of a standard deviation above the mean, not that unusual.

- Compute and interpret the Z-score for someone that scores a 19.

$$z = \frac{19-21}{6} = \frac{-2}{6} = -\frac{1}{3}$$

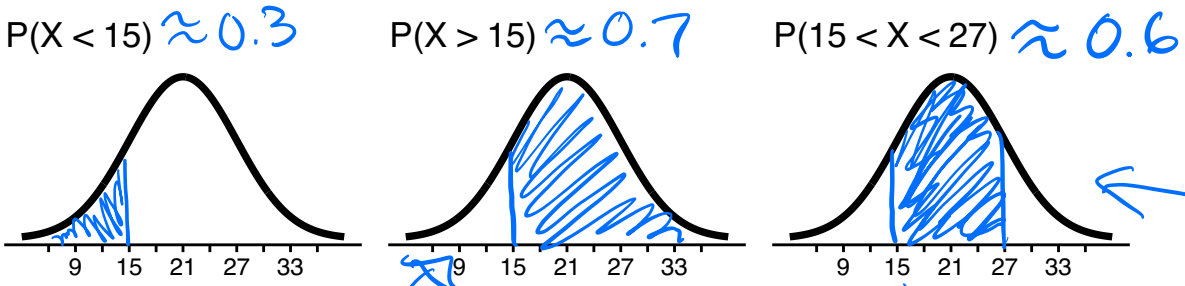
So a score of 19 is -1/3 standard deviations below the mean, so even less unusual.

- Which score is more unusual, and why?



Area under the curve = Probability

Let X represent an ACT score. Shade the area under the curve corresponding to each case, and make a guess at the probability for each case.



pnorm(15,mean=21,sd=6) = 0.1587

1-pnorm(15,mean=21,sd=6) = 0.8413

pnorm(27,mean=21,sd=6)-pnorm(15,mean=21,sd=6) = 0.6827

How do we get those areas/probabilities exactly? **We can use R for calculations!**

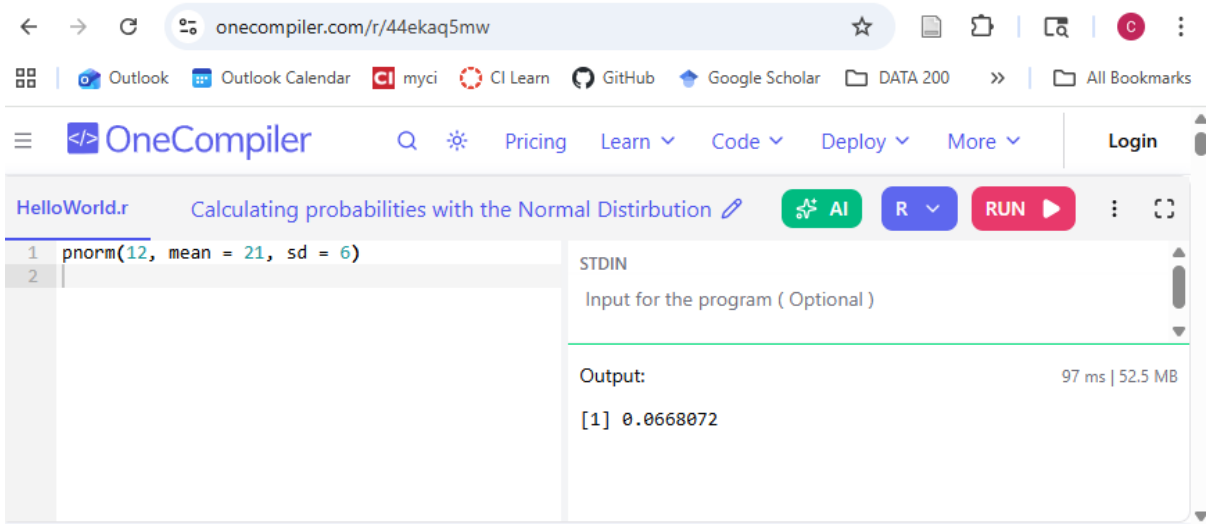
The `pnorm()` function will calculate $P(X < x)$, the probability of being less than a value.

```
pnorm(12, mean = 21, sd = 6)
```

```
[1] 0.0668072
```

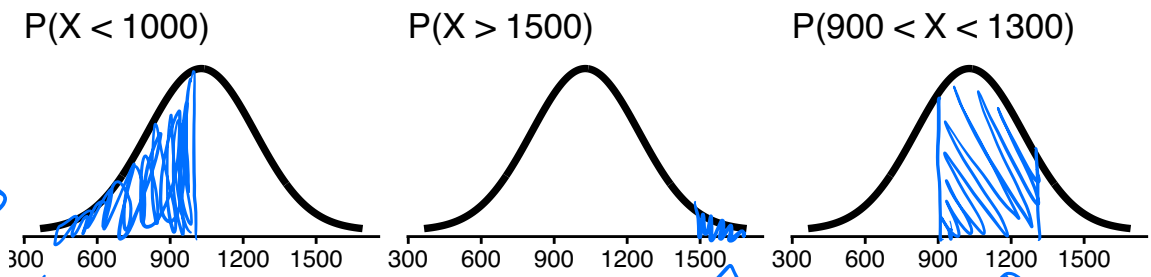
Calculate the exact probabilities you approximated above using `pnorm()`.

There are a lot of different software applications we can use to do R calculations. I recommend using OneCompiler https://onecompiler.com/r



Example: National Center of Education Statistics says that in 2023 national SAT scores had a mean of 1028 and a standard deviation of 221. Assuming that the normal distribution is a good model for SAT scores, for each scenario:

- shade the corresponding region under the curve, and
- calculate the probability and write the `pnorm()` call you used.



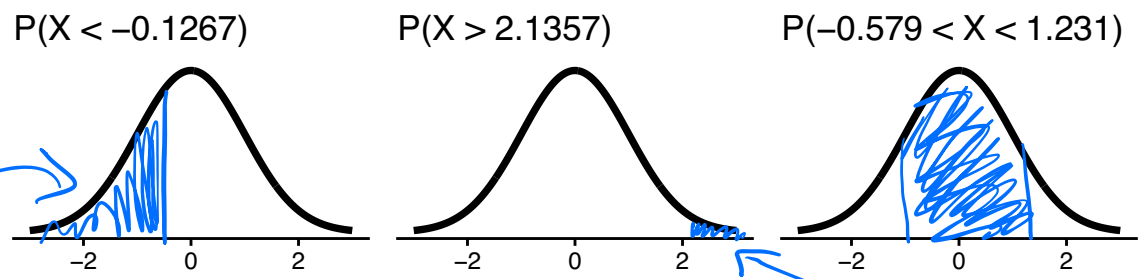
pnorm(1000, mean=1028, sd=221) = 0.4496

1 - pnorm(1500, mean=...) = 0.0164

pnorm(1300, ...) - pnorm(900, ...) = 0.6096

Example: Now assume we have a variable $X \sim N(\mu = 0, \sigma = 1)$, for each scenario:

- shade the corresponding region under the curve, and
- calculate the probability and write the `pnorm()` call you used.



pnorm(-0.1267, mean=0, sd=1) = 0.4496

1 - pnorm(2.1357, ...) = 0.0164

pnorm(1.231, ...) - pnorm(-0.579) = 0.6095

If $X \sim N(\mu, \sigma)$, then $Z \sim N(0, 1)$. We call this, $N(0, 1)$, a **standard normal**.

We've learned if we know μ and σ for a variable with a normal distribution,

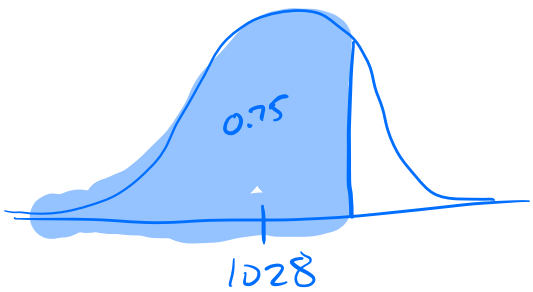
- we can calculate probabilities given values x, using `pnorm()`,
- we can also calculate values x, given probabilities, using `qnrom()`!

Percentiles are the percent of observations that fall below a given value. For example, the median is the 50th percentile.

Example: Suppose I take the SAT and score at the 75th percentile, what is my score? What percent of people scored higher than me?

If X = SAT score

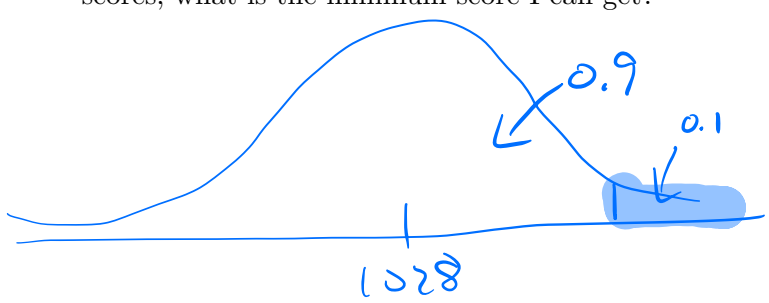
$X \sim N(1028, 221)$



qnorm(0.75, mean=1028, sd=221) = 1177.062

So, if I score at the 75th percentile that means my score is 1177 and 75% of people scored worse than me.

Example: Suppose I'm extra competitive and want to score in the top 10% of all SAT scores, what is the minimum score I can get?



qnorm(1 − 0.1, mean=1028, sd=221) = $\boxed{1311}$

need the "1−" because pnorm and qnorm will always assume your working with the probability to the left.